Customer Shopping Behavior Analysis

Insights from 3,900 purchases across product categories

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Dataset Overview

3.9K

Transactions

Purchase records analyzed

18

Features

Data points per transaction

37

Missing Values

Review ratings imputed

Demographics, purchase details, and shopping behavior

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Python Data Preparation

Data Loading

Imported with `pandas`, checked structure with `df.info()`

Cleaning

Imputed missing ratings using median by category

Standardization

Renamed columns to snake case for readability

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Feature Engineering

Age Groups

Binned customer ages into segments

Purchase Frequency

Calculated days between purchases

Database Load

Connected to PostgreSQL for SQL analysis

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SQL Analysis: Revenue Insights

Gender Revenue

- Male: 75191
- Female: 157890

Age Group Revenue

- Young Adult: 62143
- Middle-aged: 59197
- Adult: 55978
- Senior: 55763

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Product Performance Top 1 per cetagory

Jewelry

171 orders in Accessories

Blouse

171 orders in Clothing

Sandals

160 orders in Footwear

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Top-Rated Products

1

Gloves

3.86 average rating

2

Sandals

3.84 average rating

3

Boots

3.82 average rating

4

Hat

3.80 average rating

5

Skirt

3.78 average rating

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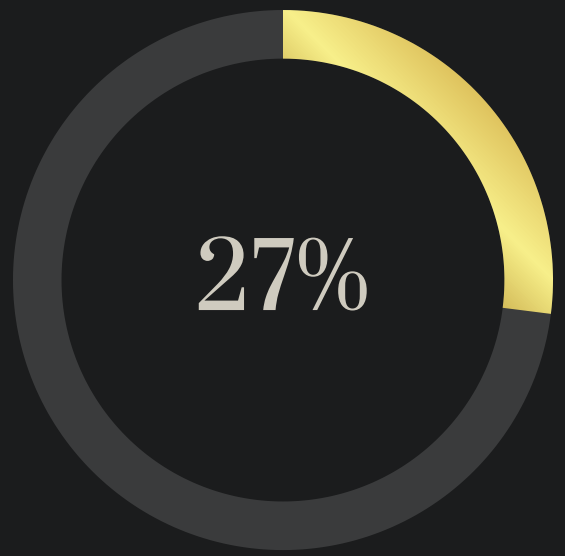
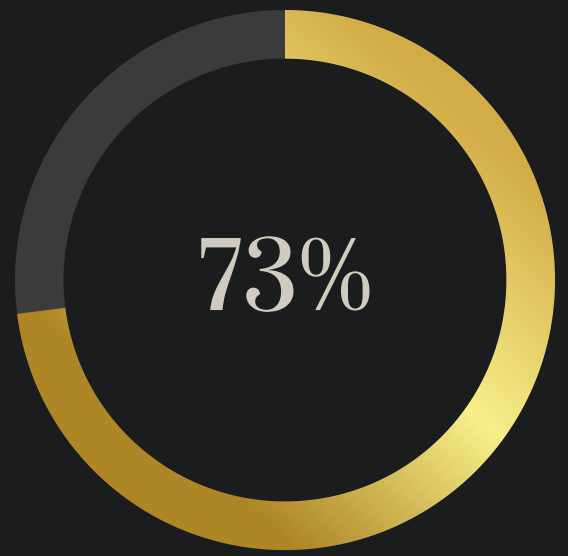
Subscription Analysis



Subscribers

1,053 customers subscribed



Non-Subscribers

2,847 customers not subscribed

Subscribers

Avg: $59.49 | Total: $62,645

Non-Subscribers

Avg: $59.87 | Total: $170,436

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Shipping & Discounts

Express Shipping

60.48 average purchase

Standard Shipping

58.46 average purchase

Discount-Dependent Products

- Hat: 50.00%
- Sneakers: 49.66%
- Coat: 49.07%
- Sweater: 48.17%
- Pants: 47.37%

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Business Recommendations

01

Boost Subscriptions

Promote exclusive benefits

02

Customer Loyalty Programs

Reward repeat buyers

03

Review Discount Policy

Balance sales with margins

04

Product Positioning

Highlight top-rated items

05

Targeted Marketing

Focus on high-revenue groups

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